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RAILWAY DEFECT FALSE ALARM MITIGATION BY MULTIPLE TRACK CONFIGURATION ANALYSIS

Abstract

A track defect false alarm mitigation apparatus for a rail vehicle that is conveyed on a railway includes a processor onboard the rail vehicle that is constructed to perform feature detection on a track image captured from the railway to label an attribute in the track image as a feature of the railway captured in the track image. Model parameters for a track detection model are accepted from another processor. A track detection model is executed using the model parameters provided thereto to generate a track configuration context that predicts the locations of track configuration components. An attempt is made to register the feature and the track configuration context one with the other. The issuance of an alert of a track defect is excluded except in response to the feature registering with the track configuration context to a confidence level.

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Background/Summary

TECHNICAL FIELD

[0001] Generally, the present disclosure relates to false alarm mitigation for rail vehicles such as railway rail vehicles. More specifically, the subject matter disclosed herein is directed to railway track defect false alarm mitigation using multiple track configuration analysis techniques including computer vision (CV) and artificial intelligence (AI).

BACKGROUND

[0002] Railway track defects pose not only a hazard to rail vehicles, but also to persons aboard and surrounding the rail vehicle. Accordingly, railway operators deploy different systems and methods to detect such track defects as track buckles and other track deformation that permit a rail vehicle other than constrained travel thus increasing the likelihood of an accident.

[0003] US Patent Application Publication 2021/0078622 is directed to a system for detecting buckled rail in a railroad track. A forward-facing camera is mounted on a locomotive of a rail vehicle traveling on the railroad track. The forward-facing camera may be configured to capture images of the rails ahead of the locomotive, to detect a buckled rail in the images, and to measure dimensions of the buckled rail. Additionally, an event recorder onboard the locomotive is in communication with the forward-facing camera to receive data therefrom that are associated with the buckled rail when the forward-facing camera detects the buckled rail. The data may include images of the buckled rail and the dimensions of the buckled rail. The event recorder may be further configured to transmit an alarm signal to a display interface of a remote unit if the dimensions of the buckled rail meet a predetermined threshold.

[0004] While existing systems such as those described above offer improved defect detection than ever before, the number of false alarms can become a burden on the operator in that each alarm, whether false or not, must be attended to. Accordingly, research and engineering resources are being devoted to mitigating false alarms to minimize operator tedium.

SUMMARY

[0005] In one aspect of the present inventive concept, a track defect false alarm mitigation apparatus for a rail vehicle that is conveyed on a railway includes a processor onboard the rail vehicle that is constructed to perform feature detection on a track image captured from the railway to label an attribute in the track image as a feature of the railway captured in the track image. Model parameters for a track detection model are accepted from another processor. A track configuration context, as defined below, is generated by the track detection model using model parameters provided thereto by the other processor. The track configuration context predicts the locations of track configuration components. An attempt is made to register the feature and the track configuration context on the the other. An alert of a track defect may be issued except in response to the feature and the track configuration context aligning to a confidence level.

[0006] In another aspect of the present inventive concept, a railway enterprise system of rail vehicles having an onboard processor onboard each of the rail vehicles and an offboard processor physically removed from and communicatively coupled to the onboard processor. Each of the onboard processors accepting a track image captured from the railway by an imaging device.

Model parameters for a track detection model are accepted from another processor. Feature detection is performed on the track image to label an attribute in the track image as a feature of the railway captured in the track image. A track configuration context is generated using model parameters provided thereto. The track configuration context predicts locations of track configuration components based on the track image. An attempt is made to register the feature with the predicted track configuration components based on the track image. An alert is issued except in response to the feature registering with the track configuration to a confidence level. The offboard processor is constructed to train the offboard track detection model on the pixels representing the known track configurations to meet a cost threshold and to convey parameters of the trained offboard track detection model to the onboard processor as the parameters of the track detection model accepted thereat.

[0007] In yet another aspect of the present inventive concept, a track defect false alarm mitigation method for a rail vehicle that is conveyed on a railway includes performing, at a processor onboard the rail vehicle, feature detection on a track image captured from the railway to label an attribute therein as a feature of the railway. The processor onboard the rail vehicle accepts model parameters for a track detection model from another processor. A track configuration context is generated by artificial intelligence using the track detection model configured with the model parameters provided thereto by the other processor. The processor onboard the rail vehicle attempts to register the feature with the track configuration context. An alert of a track defect is issued except in response to the feature registering with the track configuration context to a confidence level.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic block diagram of an exemplary rail system on which rail vehicles are conveyed in an embodiment of the present inventive concept.

[0009] FIGS. 2A-2C are graphic representations of select operational states of an exemplary feature detection and false alarm mitigation process by which the present inventive concept can be embodied.

[0010] FIG. 3 is a schematic block diagram of an exemplary arbitration and false alarm mitigation processor by which the present inventive concept can be embodied.

[0011] FIG. 4 is a schematic block diagram of an exemplary onboard track defect handling processor by which the present inventive concept can be embodied.

[0012] FIG. 5 is a schematic block diagram of an exemplary offboard processor by which the present inventive concept can be embodied.

[0013] FIG. 6 is a flow diagram of an exemplary false alarm mitigation process by which the present inventive concept can be embodied.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0014] The present inventive concept is best described through certain embodiments thereof, which are described in detail herein with reference to the accompanying drawings, wherein like reference numerals refer to like features throughout. It is to be understood that the term invention, when used herein, is intended to connote the inventive concept underlying the embodiments described below and not merely the embodiments themselves. It is to be understood further that the general inventive concept is not limited to the illustrative embodiments described below and the following descriptions should be read in such light.

[0015] Additionally, the word exemplary is used herein to mean, “serving as an example, instance or illustration.” Any embodiment of construction, process, design, technique, etc., designated herein as exemplary is not necessarily to be construed as preferred or advantageous over other such embodiments.

[0016] The figures described herein include schematic block diagrams illustrating various interoperating functional modules. Such diagrams are not intended to serve as electrical schematics and interconnections illustrated are intended to depict signal flow, various interoperations between functional components and/or processes and are not necessarily direct electrical connections between such components. Moreover, the functionality illustrated and described via separate components need not be distributed as shown, and the discrete blocks in the diagrams are not necessarily intended to depict discrete electrical components.

[0017] The techniques described herein are directed to false alarm mitigation of railway defects. Upon review of this disclosure and appreciation of the concepts disclosed herein, the ordinarily skilled artisan will recognize other false alarm mitigation contexts in which the present inventive concept can be applied. The scope of the present invention is intended to encompass all such alternative implementations.

[0018] As used herein, an “attribute” in an image is intended to refer to a “quantifiable characteristic of a phenomenon present in the image over a set of pixels.” A “feature,” as the term is used herein, is “an attribute (as defined above) associated with a label in memory circuitry,” and “label” is intended to mean “computer-readable data (e.g., metadata), including those descriptive of the feature (e.g., feature name), that is conditionally produced by a subject matter expert (SME) and/or computer-executable deep learning process that has been trained on the quantifiable characteristic being the feature and that has had the image data provided as input thereto.” Appropriately, when used as a verb, “label” is intended to mean, “associating, in the memory circuitry, the descriptive data of the feature with the quantifiable characteristic of the phenomenon present in the image.” A “track configuration context” is intended to refer to a data set that indicates placement of the railway components predicted by an artificial intelligence model based on the track image under scrutiny. It is to be understood that these designations are not intended to substitute for the ordinary and customary meanings thereof as used in the artificial intelligence arts and, indeed, are intended to be substantially consistent therewith.

[0019] FIG. 1 is a schematic block diagram of an exemplary enterprise **100** that operates a fleet **15** of rail vehicles **11a-11n**, representatively referred to herein as rail vehicle(s) **11**, each under the power of a locomotive **10**, although practicing the present inventive concept is not limited to such. Operational control over a fleet management infrastructure **50** may also fall under the purview of enterprise **100**, while railway **20** may be the responsibility of a third party. It is to be understood, however, that the present inventive concept is applicable to other machines that ride on rails and similarly constrained pathways without departing from the spirit and intended scope of the present inventive concept.

[0020] Certain railway defects, e.g., track buckling, may be identified through computer-readable images such as those captured by exemplary camera **105**. Camera **105** may be constructed or otherwise configured to generate digital images from captured image fields **25** of railway **20**. Each image field **25** may be bounded by an image capture diameter D that is defined by the camera optical aperture angle θ . It is to be understood, however, that image field **25** may not necessarily be circular, but may be elliptical with the major axis aligned parallel to railway **20**. Camera **105** may be mounted on locomotive **10** to position image field **25** downrange by a distance R as it is being conveyed along railway **20** with locomotive **10**. As an example, the field of view may capture the nose of locomotive **10** and a region of track around 33° downrange of locomotive **10**. In cases where locomotive **10** is not the leading car of the rail vehicle, camera **105** may be mounted on that leading car. However, it is to be understood that camera **105** may be installed anywhere on rail vehicle **11** but is preferably mounted to ensure an unobstructed view of the relevant portions of railway **20**. The distance R may be selected according to a combined response time of the machine operator and the digital processing time of track feature detection, classification, localization, measurement, presentation, etc., when rail vehicle velocities are considered.

[0021] As illustrated in FIG. 1, each rail vehicle **11** in fleet **15** may be outfitted with an onboard

processor **110** that is communicatively coupled to an offboard processor **150** through a communications channel **35**. Communications channel **35** may be constructed from, for example, a corresponding communications link **30a-30n** and a communications link **32** coupled one to another according to signaling interfaces and data transfer protocols of a telecommunications network **70**. To implement such coupling, onboard processor **110** and offboard processor **150** may include communications components **126** and **152**, respectively, that are each constructed or otherwise configured to synchronize, modulate/demodulate, code/decode, etc., radio signals conveying data on communication channel **35**. The present inventive concept may be embodied in various signal interfaces, data transfer protocols, etc., known to skilled artisans without departing from the spirit and intended scope thereof.

[0022] Onboard processor **110** may be constructed or otherwise configured with computer readable instructions and/or electronic processing and memory circuitry to perform various functions, each depicted in FIG. **1** as a distinct processor component solely for purposes of explanation. As will be discussed further below, offboard processor **150** may be similarly outfitted and may be realized on data processing resources of fleet management infrastructure **50**. Any of the exemplary processor components of onboard processor **110** and offboard processor **150** may be implemented as, for example, microprocessors, graphics processors, artificial intelligence processors, volatile and/or nonvolatile memory components, programmable logic such as field programmable logic arrays (FPGAs) etc., that are sufficient for artificial intelligence, e.g., deep learning track detection, classification, localization and measurement of features identified by machine vision in the track images, such as those described below. Further, certain circuits may extend across the depicted processor components and be shared among processes executing thereon.

[0023] Exemplary imaging/image processor **112** may be constructed or otherwise configured to optically capture track characteristics from image field **25** and to electronically digitize, pixelate, format, etc., those track characteristics into image data that are sufficient for both machine vision and deep learning processing. Such image data is referred to herein as a track image although it is to be understood that more than tracks may be captured and contained in the image data.

Imaging/image processor **112** may produce one or both of static images and full motion video at high resolution, e.g., at least 4K ultra-high definition (UHD; 3840×2160 pixels per frame). It should be inferred that the circuitry handling such high resolution data is present in the disclosed embodiments of the present inventive concept despite explicit details thereof being omitted from this disclosure. Skilled artisans will recognize various circuits that can be used in this regard without departing from the spirit and intended scope of the present inventive concept.

[0024] Feature detection processor **116** may be constructed or otherwise configured to perform various image processing tasks in furtherance of segregating objects of interest, e.g., track rails, track ties, track bed boundaries, and other railway components, from background elements and objects of lesser interest in image field **25**. Railways are generally realized in regular, patterned structures having well defined edges. Accordingly, feature detection processor **116** may apply one or more filters that, for example, enhance pixel luminance of high gradient edges and/or subdue the pixel luminance of low gradient background regions. The present inventive concept may be practiced using various image filters that realize these results; however, it is to be understood that other feature detection/extraction techniques, including nonfilter implementations, pattern recognition, and the like.

[0025] Feature detection processor **116** may apply a curve fitting technique to the highlighted (e.g., luminance enhanced) attributes. Such a curve may enclose a region of pixels that corresponds in shape to a component of the track configuration under scrutiny. As used herein, the term “track configuration” is intended to refer to an assemblage of railway components that achieves a specific railway function, e.g., switches, crossovers, straight and curved runs, etc. Once feature detection processor **116** has computed such a curve, its characteristics may be provided to a requesting process along with a confidence measure or value that the computed curve fits the enhanced pixel

data. Feature detection processor **116** may apply labels that have been selected in accordance with, for example, known component shapes and other known track configuration data, auxiliary data, e.g., satellite geopositioning data, and known geoposition of railway components. A confidence measure may be computed as to whether an attribute in a track image is indeed a feature of the railway. For example, the curve fitting process may find a set of polynomial curves that pass through a set of nodes, representatively illustrated at node **215**, arranged along and/or over highlighted attributes. A least squares linear regression technique, for example, may be employed that, as part of its optimization processing, provides a confidence measure and/or uncertainty quantity as to how well the curve follows the highlighted pixel data and corresponding physical characteristic of the corresponding feature. The confidence value may be used to determine whether anomalies exist in the track image that should be investigated. For example, higher order terms in a proposed curve fit may indicate other than the regular shapes of a railway. Embodiments of the present inventive concept may determine whether the higher order curve is indeed part of the railway or indicates another anomaly that might be useful, such as for training. Applying a curve fitting technique through such attributes may result in the confidence level generated by the curve fitting method being lower than without the erroneously included attribute data, often substantially so. Unless the confidence measure meets a confidence threshold condition, e.g., greater than or equal to 85% confidence, the potential track defect may undergo additional analysis prior to a track defect alarm being issued to warn the rail vehicle operator. The present inventive concept is not limited to the manner in which confidence in a data fitting technique is computed so long as it has a scale on which confidence thresholds may be set as exemplified herein.

[0026] Onboard processor **110** may include a validation processor **118** by which it is determined whether attributes and/or features present in a track image are indeed a feature of a track configuration present in the track image. To do so, embodiments of the present inventive concept may realize an onboard DL model **119** that is constructed or otherwise configured to define a track configuration context against which attributes and/or features present in the track image are analyzed. For example, if an attribute is not within the track configuration context, it is identified as being outside the track configuration and therefore an anomaly. This aspect of embodiments of the present inventive concept is described further below.

[0027] False alarm mitigation processor **122** may be constructed or otherwise configured to determine whether the attribute in the track image captured from the railway is a feature of the railway captured in the track image. Unless it is determined that the attribute in the image is not a feature of a track configuration that conforms to a physical specification, e.g., the attribute remains unlabeled and the DL confidence measure meets an alarm confidence condition, false alarm mitigation processor **122** may exclude, suppress or make otherwise easily disregarded the issuance of the track defect alarm. Should the alarm confidence condition be met, false alarm mitigation processor **122** may issue the track defect alarm and may concurrently route the subject track image in which the attribute is presented to offboard processor **150** for additional processing.

[0028] DL model update processor **124** may be constructed or otherwise configured to populate DL track detection model **119** with model parameters, e.g., weights on network edges, obtained from offboard processor **150**. DL track detection model **119** may be configured identically to offboard DL track detection model **157** executing on DL track detection processor **156**. DL track detection model **119** may be trained remotely (i.e., offboard from the rail vehicle **11**); instead, offboard DL track detection model **157** may be trained and the parameters (e.g., edge weights) achieved when DL track detection model **157** has met a cost function requirement are transferred to onboard DL track detection model **119** executing on DL track detection processor **118** through DL model update processor **124**.

[0029] Similarly with onboard processor **110** described above, offboard processor **150** may be constructed or otherwise configured with computer readable instructions and/or electronic processing and memory circuitry to perform various functions, each depicted in FIG. **1** as a distinct

processor component solely for purposes of explanation.

[0030] Data aggregation/image pooling component **154** may be constructed or otherwise configured to accept image and other data from rail vehicles **11** in fleet **15** and to assign each datum a storage location in memory. As is discussed further below, memory circuitry of offboard processor **150** may be partitioned into an onboard image pool storing unlabeled or partially labeled track image data for further processing and/or analysis and a labeled image pool storing labeled track image data that may be used for training DL track detection model **157** executing on DL track detection processor **156**.

[0031] DL track detection, classification, localization and measurement processor **156**, referred to herein simply as DL track detection processor **156**, may be constructed or otherwise configured to accept pooled images and aggregated data and to cooperate with DL training processor **158** to train its underlying DL track detection model **157** with labeled track images provided by DL training processor **158**. DL training processor **158** may be constructed or otherwise configured to select training images from a labeled image pool, modifying DL track detection model parameters, e.g., edge weights, based on an evaluation against a cost function, and terminating the modification of the DL track detection model parameters in response to convergence of the cost function (e.g., minimization, maximization) to a convergence threshold. The training image set may include images of track configurations that meet physical specifications and that have been previously encountered by fleet **15**, captured in digital image(s) thereby and labeled to identify the captured railway features. If convergence of the cost function cannot be achieved to the convergence threshold, DL training processor **158** may provide the track image under scrutiny to image routing processor **162** for further dispensation. As with validation processor **118**, input to offboard DL track detection processor **156** may be the track image under scrutiny and the output thereof may be, among other things, a track configuration context associated with a probability that attributes in the track image are railway features.

[0032] As indicated above, image routing processor **162** may be constructed or otherwise configured to route track images for associating labels with relevant image attributes thereby forming respective labeled features. For example, when offboard DL track detection processor **156** indicates a high confidence match with a known track configuration on which offboard DL track detection model **157** has been trained, image routing processor **162** may provide the subject track image and the labels associated therewith, to the extent that such have been generated for the attribute under scrutiny, to an auto-labeling process that generates labels and associates those labels with the attribute under scrutiny in the track image. This may be achieved by analyzing various cues in the context of the attribute in the track image and choosing a label with the highest likelihood of being correct. On the other hand, when offboard DL track detection processor **156** indicates a low confidence match with a known track configuration on which offboard DL track detection model **157** has been trained, image routing processor **162** may provide the subject track image and the labels associated therewith, to the extent that such have been generated for the attribute under scrutiny, to manual labeling routing process. Manual labeling routing may provide the track image under scrutiny and associated data to subject matter experts (SMEs) for manual labeling and storing the manually labeled track image to the labeled image pool where it can be used for DL model training.

[0033] FIGS. 2A-2C, representatively referred to herein as FIG. 2, are graphic representations of select operational states **210-230** of an exemplary feature detection and false alarm mitigation process **200** performed on track image **205**. Here, the term “operational state” is intended to refer to a point in the track defect false alarm mitigation processing after one operation has completed but before the temporally next operation starts. It is to be understood, however, that the inventive concept described herein is not limited to temporally sequential data processing.

[0034] At operational state **210**, process **200** may have applied a gradient filter, such as an edge detection filter, to track image **205** as well as performed a curve fitting process on the detected

edges. Computationally, the curve fit finds the best polynomial fit, for example, that passes through a set of nodes, representatively illustrated at node **215**, arranged on the subject attribute. Certain embodiments may employ a least squares linear regression technique that provides a confidence measure and/or uncertainty quantity as part of its optimization processing. The attribute(s) may be highlighted through edge detection, for example, and may have labels associated with features that conform to the polynomial fit associated with the conditions under which track image **205** was taken. As is illustrated in the figure, in certain of those conditions, primary feature detection processor **116** may detect image attributes having high luminescent gradient, for example, that are not part of railway **20**, such as portions of an adjacent fence **217**. Moreover, feature detection processor **116** may attempt to apply curve fitting through such attributes, as representatively illustrated at node **215'**. In doing so, higher order terms in the curve fit polynomial may be generated which may be indicative of an attribute that is not part of the railway.

[0035] As discussed above, feature detection processor **116** may identify certain track features from the manner in which the curve fits the attribute data. At operational state **210**, the relative locations of nodes **215** and knowledge of track configurations may allow identification of rails, representatively illustrated at rail label **214**, rail ties, representatively illustrated at tie label **213**, among other railway components. Additionally, feature detection processor **116** may be unable to identify elements of fence **217** from attempting to curve fit through nodes **215** and nodes **215'** concurrently. Consequently, the confidence measure computed through, for example, a least squares technique, may fail to meet a confidence threshold.

[0036] At operational state **220**, onboard DL model **119** has been executed that may have been trained on track configurations that conform to product physical specifications to predict the track configuration that fits the input data, e.g., track image **205**. As illustrated at operational state **220**, onboard DL model **119** has predicted a track configuration context **225** in which ties **222**, rails **224** and track bed **226** are contained. As illustrated in the figure, however, edge detected attributes corresponding to nodes **215'** are not predicted by DL model **119** in that they fall outside the track configuration context **225** predicted by validation processor **118**.

[0037] At operational state **230**, a new feature search may have been performed with new search parameters that are confined to search within context **225**. The result of the renewed search is nodes **215'** in region **232** being excluded. That is, only those attributes that fall within the boundaries of track configuration context **225** are located in the search. When nodes **215'** are excluded as a result of falling outside track configuration context **225**, the confidence measure may increase to indicate that nodes labeled X in region **236** are not part of railway **20**. It is to be noted that left and right bed boundaries **234l** and **234r**, left and right rails **235l** and **235r** and rail ties **238a-238m** have been isolated from attributes of less interest.

[0038] FIG. **3** is a schematic block diagram of an exemplary track defect false alarm mitigation (TDFAM) apparatus **300** by which the present general inventive concept can be embodied. Whereas TDFAM apparatus **300** may receive offboard support, such as by way of offboard training **336**, the functionality thereof may be constructed, programmed and/or otherwise configured to execute onboard each rail vehicle **11** autonomously. Skilled artisans will recognize well-known techniques by which such functionality may be achieved in view of the descriptions herein.

[0039] As depicted in the example, track images **302** may be provided to both onboard feature detection processing **325** and onboard validation processing **332**. Onboard feature detection processing **325** may be configured as the primary feature detection technique while onboard validation processing **332** may generate a track configuration context that specifies the placement of components within a track configuration according to physical specifications thereof. Onboard validation processing **342** may be generated by, for example, an AI predictor or estimator, e.g., AI executive **334**, executing a track configuration model. The example embodiments described herein implement DL as the AI paradigm, but such is not a requirement to practice the present inventive concept. Onboard feature detection processing **325** may be implemented using known computer

vision techniques such as gradient filtering **304**, feature discovery **304** on shapes highlighted by gradient filtering **304** and attribute analysis **316** that resolve features (labeled attributes) from attributes in the track image under scrutiny to generate feature(s) and attribute(s) data **322**. Feature discovery processing **308** may search for graphical elements having track component shapes according to a set of search rules **306** prior to attribute analysis **316**, which may compare those graphical elements against a specification **314** for determining whether an attribute meets physical dimensions for components of a track configuration. Together, gradient filter **304**, feature discovery processing **308** and attribute analysis **316**, illustrated at onboard feature detection processor **325**, may be constructed or otherwise configured to generate features/attributes **322**, where the features comprise attributes that have been labeled.

[0040] As illustrated in FIG. **3**, features/attributes **322** and track configuration context estimation **338** may be provided to feature registration and comparison processing **342**. Here, the term “registration” is used in its customary sense, i.e., a condition of correct alignment or proper relative position. In the present case, the registration may attempt to align the detected features with features predicted by onboard validation processing **332**. The alignment allows a comparison that determines how well the data are registered, e.g., the detected features are within the predicted track configuration context. When there is high confidence that features/attributes **322** are aligned with track configuration context estimation **338**, high confidence rail detection may be performed to isolate or highlight the rails of the railway. Rail defect detection **346** may be performed to determine whether an anomaly in the highlighted rails exists. If so, a track defect alarm **348** that might otherwise be activated is suppressed. When no anomaly exists, the aforementioned alarm may be issued to the rail vehicle operator. On the other hand, if there is low confidence as determined by feature registration and comparison processing **342**, context limiter processing **318** may format modified rules **312** that may limit the search for track configuration features performed by feature discovery processing **308** to within the predicted track configuration context. The results of the new search may be passed feature discovery processing **308** and the process repeats with the modified search rules **312**.

[0041] FIG. **4** is a schematic block diagram of an exemplary onboard track defect handling processor **400** by which the present inventive concept can be embodied. Onboard track defect handling processor **400** may implement much of the functionality described above with reference to onboard processor **110**. An optical imager **405** may be constructed or otherwise configured to capture light and generate electrical signals appropriately and an image processor **410** that may be constructed or otherwise configured to assemble a track image from which features can be recognized by machine. Optical imager **405** and image processor **410** may be implemented as, for example, the UHD video camera described above with reference to imaging/image processor **112**. Track images may be provided to rail and track feature detection processor **415**, which may be constructed or otherwise configured to label image attributes as railway features that the attributes, once recognized, represent. Labeled track data may be provided to rail detection, localization and measurement processor **420** by which rails are isolated or otherwise highlighted in the track image. Detected rails and context data provided by a DL rail and infrastructure detection, classification, localization and measurement processor **445** may be registered one with the other to determine whether a track defect exists, as described herein, which may be performed by track defect detection, classification, localization, measurement and visualization processor **425**. Such track defects, when detected, are reported and a track defect alarm may be issued through defect alarm/reporting processor **430**. These data may be provided to transmitter **440** for conveyance to offboard processing resources depicted in FIG. **5**.

[0042] As stated above, the DL model onboard rail vehicle(s) **11** may be a duplicate of a DL model offboard rail vehicle(s) **11** that has been trained thereat. Such model training performed offboard may be characterized by its edge weights, for example, which may be conveyed through receiver **455** to DL model update processor **450** communicatively coupled thereto. The onboard track

configuration model may execute under DL rail and infrastructure detection, classification, localization and measurement processor **445**.

[0043] FIG. 5 is a schematic block diagram of an exemplary offboard processor **500** by which the present inventive concept can be embodied. Offboard processor **500** may implement much of the functionality described above with reference to offboard processor **150**. Track defect alarms are analyzed and provided to rail vehicles **505** as intelligence supporting feature detection and the like. A data ingest processor **510** may receive image, labeled features, classification, statistical, etc., data from the various rail vehicles **505** and store these data to memory circuitry **515** at onboard unlabeled image pool **530**. Partially or completely unlabeled images stored in onboard image pool **525** may be provided to a subject matter expert labeling workflow **570** by which features are manually labeled and returned into image pool **530** as labeled track image data.

[0044] As illustrated in FIG. 5, labeled images may be retrieved from image pool **585** and provided to an offboard DL track detection model update processor **545** that updates DL model/network **542** executing on DL track detection processor **540**. Once the DL network has been trained, DL track detection model definition data **550**, e.g., edge weights, may be transferred to DL model update processor **450** through receiver **455**. Certain federated learning techniques may be used to embody the present inventive concept in which a single model is trained, and the results distributed among entities, e.g., rail vehicles **11**, at which the model is used for its regular intended purpose.

[0045] Once the model of DL track detection processor **545** has been trained, image data from image pool **525** may be provided thereto and analyzed. Where there is low confidence in a match between a feature and the track configuration context, the track image data **560** may be provided to SME labeling workflow **570** where images of interest are identified for manual labeling through a subject matter expert pool **99**. In so doing, the volume of manual labeling work by subject matter experts is limited to special cases.

[0046] Upon DL track detection processor **545** determining that image label data associated image data matches the corresponding track configuration with high confidence, the high confidence image data **565** may be provided to an auto-label processor **580** by which known labels are associated with detected features. In certain embodiments, auto-labeled track image data is conveyed through SME labeling workflow **570** for verification. Track image data that has undergone manual labeling as well as the track image data that has undergone auto-labeling may be stored in labeled image pool **585** as labeled image data and used for DL model training with other such labeled data.

[0047] FIG. 6 is a flow diagram of an exemplary false alarm mitigation process **600** by which the present inventive concept can be embodied. False alarm mitigation process **600** may be implemented as interoperating processes, e.g., an onboard process **610** and an offboard process **650**. Exemplary onboard process **610** may be executed on onboard processor **110** to analyze track images for potential track defects and to mitigate, with the support of deep learning, the number of visual track inspections the machine operator must perform to verify that a given track defect alarm is not due to irrelevant image attributes and/or artifacts, i.e., false track defect alarms. Exemplary offboard process **650** may be executed on offboard processor **150** to label features in the track images that were not identified by onboard process **610** and to train the deep learning model using the labeled track images.

[0048] As is illustrated FIG. 6, track images may be acquired in operation **615**, such as by an onboard camera. In operation **620**, an AI validation model **622** may be executed on the track images and in operation **625**, primary feature detection may be performed on the track images. In operation **630**, an attempt is made to register feature detection data onto artificial intelligence track configuration data or, stated another way, by way of a successful registration attempt to a registration confidence level, pixels corresponding to the features detected in operation **625** fall within a track configuration context generated in operation **620**. When registration is successful, as determined in operation **635**, process **600** may transition to operation **640** by which high

confidence rail detection is performed. Height confidence track configuration detection may highlight pixels corresponding to the rails of the railway. Process **600** may then transition to operation **645** by which rail defects are detected. This may be achieved on the highlighted rails by determining whether the highlighted pixels are arranged in accordance with railway specifications. If a defect is detected, as determined in operation **650**, process **600** may transition to operation **655** by which a track defect alarm is issued to at least the rail vehicle's conductor. In operation **669**, the track image is conveyed to the offboard processor and process **600** may continue at operation **615**, by which a next track image is acquired.

[0049] If it is determined that feature detection data and artificial intelligence context data do not align, as determined in operation **635**, process **600** may transition to operation **660**, by which the feature detection search is reinitialized to search within the context bounds generated via onboard model **622**. Such feature detection may be realized by operation **663**. In operation **665**, it is determined once again whether the feature detection data and artificial intelligence context data align. If so, process **600** may transition to operation **640** by which high confidence rail detection is once again performed. If the feature detection data and the artificial intelligence modeled data do not align, the image is ignored in operation **667** by which the track image under scrutiny is ignored and process **600** may transition back to operation **615** to analyze the next track image. Offboard process **670** may accept unlabeled (including partially labeled) track images from fleet rail vehicles **15** and store those track images in unlabeled image pool **675**. In operation **680**, an AI validation model, such as a DL model, may be executed, the results from which being processed in accordance with the confidence level in the solution: high confidence feature matches may be processed by an auto-labeling method in operation **685** and low confidence feature matches being routed through the manual labeling workflow in operation **690**. In either case, the resulting labeled image may be stored in labeled image pool **695** to be used in model training in operation **697**. The newly trained model may be validated in operation **693**. The AI model, or the defining weights thereof, may be provided to update the onboard AI validation model **622** in operation **690** for all rail vehicles in fleet **15**.

[0050] Certain embodiments of the present general inventive concept provide for the functional components to be manufactured, transported, marketed and/or sold as processor instructions encoded on computer-readable media. The present general inventive concept, when so embodied, can be practiced regardless of the processing platform on which the processor instructions are executed and regardless of the manner by which the processor instructions are encoded on the computer-readable medium.

[0051] It is to be understood that the computer-readable medium described above may be any non-transitory medium on which the instructions may be encoded and then subsequently retrieved, decoded and executed by a processor, including electrical, magnetic and optical storage devices. Examples of non-transitory computer-readable recording media include, but not limited to, read-only memory (ROM), random-access memory (RAM), and other electrical storage; CD-ROM, DVD, and other optical storage; and magnetic tape, floppy disks, hard disks and other magnetic storage. The processor instructions may be derived from algorithmic constructions in various programming languages that realize the present general inventive concept as exemplified by the embodiments described above.

INDUSTRIAL APPLICABILITY

[0052] In general, the teachings of the present disclosure may find applicability in railroad industries. More specifically, the teachings of the present disclosure may be applicable to railroad industries in which the railroad tracks may be at risk of buckling. The teachings of this patent apply to any type of rail transportation. Rail transportation is generally Passenger (Transit) and Freight.

[0053] Reducing user tedium and increasing confidence in computer executed decision-making tools are major objectives of a great number of industrial applications and various embodiments of the present inventive concept meet those objectives in track defect detection for track-faring safety.

As discussed above in the BACKGROUND section, tending to every track defect alarm to segregate false positive defect detections therefrom can be a user intensive process that diverts the user's attention from tasks that might otherwise be performed. Additionally, increased confidence in track defect detection can be had by increasing confidence in computer-made decisions that false defect alarms are truly false alarms and can be ignored by the operator of a rail vehicle.

[0054] The descriptions above are intended to illustrate possible implementations of the present inventive concept and are not restrictive. Many variations, modifications and alternatives will become apparent to the skilled artisan upon review of this disclosure. For example, components equivalent to those shown and described may be substituted therefore, elements and methods individually described may be combined, and elements described as discrete may be distributed across many components. The scope of the invention should therefore be determined not with reference to the description above, but with reference to the appended claims, along with their full range of equivalents.

Claims

1. A track defect false alarm mitigation apparatus for a rail vehicle that is conveyed on a railway, the track defect false alarm mitigation apparatus comprising: an imaging device constructed to capture a track image of the railway; a processor onboard the rail vehicle and constructed to: perform feature detection on the track image to label an attribute in the track image as a feature of the railway captured in the track image; generate a track configuration context by a track detection model, the track configuration context using artificial intelligence for predicting locations of track configuration components based on the track image; attempt registration of the detected feature with the predicted track configuration context one with the other; and issue an alert of a track defect except in response to the feature registering with the track configuration context to a confidence level.
2. The track defect false alarm mitigation apparatus of claim 1, wherein the processor is further constructed to: accept model parameters for the track detection model from another processor; and configure the track detection model with the accepted model parameters.
3. The track defect false alarm mitigation apparatus of claim 2, wherein the other processor is physically removed from and communicatively coupled to the processor onboard the rail vehicle, the other processor constructed to: train an offboard track detection model on pixels representing known track configurations to meet a cost threshold; and convey parameters of the trained offboard track detection model to the processor as the parameters of the track detection model accepted thereat.
4. The track defect false alarm mitigation apparatus of claim 3, wherein the other processor is further constructed to: route the track image that meets a high confidence threshold condition to an automated labeling component by which descriptive label data of a new component of the known track configurations are associated with the attribute exclusively of human intervention to form an automatically labeled track feature; and train the offboard track detection model with the track image that includes the automatically labeled track feature represented in the track image to meet the cost threshold.
5. The track defect false alarm mitigation apparatus of claim 3, wherein the other processor is further constructed to: route the track image that meets a low confidence threshold condition to a manual labeling workflow by which the descriptive data are associated with the attribute by human intervention to form a manually labeled track feature; and train the track detection model with the track image that includes the manually labeled track feature represented in the track image to meet the cost threshold.
6. The track defect false alarm mitigation apparatus of claim 1, wherein the processor is further constructed to modify search rules under which a search for features is performed.

7. The track defect false alarm mitigation apparatus of claim 6, wherein the processor is further constructed to repeat the feature detection on the track image using the modified search rules to label the attribute in the track image as the feature of the railway captured in the track image in response to the registration attempt failing to align the detected feature with the predicted track configuration context.

8. A railway enterprise system of rail vehicles that are conveyed on a railway comprising: an onboard processor onboard each of the rail vehicles, each being constructed to: accept a track image captured from the railway by an imaging device; accept model parameters for a track detection model from another processor; perform feature detection on the track image to label an attribute in the track image being a feature of the railway captured in the track image; generate a track configuration context by the onboard track detection model configured with the model parameters provided thereto, the track configuration context being generated by artificial intelligence for predicting locations of track configuration components based on the track image; attempt registration of the detected feature with the predicted track configuration components based on the track image; and issue an alert of a track defect except in response to the feature registering with the track configuration context to a confidence level; and an offboard processor physically removed from and communicatively coupled to the onboard processor, the offboard processor constructed to: train the offboard track detection model on pixels representing track configurations to meet a cost threshold; and convey parameters of the trained offboard track detection model to the onboard processor as the parameters of the track detection model accepted thereat.

9. The railway enterprise system of claim 8, wherein the offboard processor is further constructed to: route the track image that meets a high confidence threshold condition to an automated labeling component by which descriptive label data of a new component of the known track configurations are associated with the attribute exclusively of human intervention to form an automatically labeled track feature; and train the offboard track detection model with the track image that includes the automatically labeled track feature (SME) represented in the track image to meet the cost threshold.

10. The railway enterprise system of claim 9, wherein the offboard processor is further constructed to: route the track image that meets a low confidence threshold condition to a manual labeling workflow by which the descriptive data are associated with the attribute by human intervention to form a manually labeled track feature; and train the offboard track detection model with the track image that includes the manually labeled track feature represented in the track image to meet the cost threshold.

11. The railway enterprise system of claim 8, wherein the onboard processor is constructed to modify search rules under which a search for features is performed.

12. The railway enterprise system of claim 11, wherein the onboard processor is further constructed to repeat the feature detection on the track image using the modified search rules to label the attribute in the track image as the feature of the railway captured in the track image in response to the registration attempt failing to align the detected feature with the predicted track configuration context.

13. The railway enterprise system of claim 8, wherein the onboard processor is further constructed to search for the feature of the railway captured in the track image by: applying a computer-implemented edge detection filter to the track image that is constructed to enhance luminance of the attribute in the track image; applying a computer-implemented curve fit to the enhanced luminance in the track image; and generating the confidence measure from conformance of the curve fit to a known track configuration.

14. A track defect false alarm mitigation method (600) for a rail vehicle that is conveyed on a railway, the track defect false alarm mitigation apparatus comprising: performing, at a processor onboard the rail vehicle, feature detection on a track image captured from the railway to label an attribute therein as a feature of the railway captured therein; generating a track configuration context by a track detection model configured with model parameters that predict by artificial

intelligence locations of track configuration components based on the track image; attempting registration of the feature with the predicted track configuration context one with the other; and issuing an alert of a track defect except in response to the feature registering with track configuration context to a confidence level.

15. The track defect false alarm mitigation method of claim 14, further comprising: accepting, at the processor onboard the rail vehicle, the model parameters for the track detection model from another processor; configure the track detection model with the accepted model parameters.

16. The track defect false alarm mitigation method of claim 15, further comprising: training, by another processor physically removed from and communicatively coupled to the processor onboard the rail vehicle, an offboard track detection model on pixels representing known track configurations to meet a cost threshold; convey parameters of the trained offboard track detection model to the processor as the parameters of the track detection model accepted thereat.

17. The track defect false alarm mitigation method of claim 16, further comprising: routing the track image that meets a high confidence threshold condition to an automated labeling component by which descriptive label data of a new component of the known track configurations are associated with the attribute exclusively of human intervention to form an automatically labeled track feature; and training the offboard track detection model with the track image that includes the automatically labeled track feature represented in the track image to meet the cost threshold.

18. The track defect false alarm mitigation method of claim 17, further comprising: routing the track image that meets a low confidence threshold condition to a manual labeling workflow by which the descriptive data are associated with the attribute by human intervention to form a manually labeled track feature; and training the track detection model with the track image that includes the manually labeled track feature represented in the track image to meet the cost threshold.

19. The track defect false alarm mitigation method of claim 18, further comprising: modifying search rules under which a search for track features is performed by the processor.

20. The track defect false alarm mitigation method of claim 19, further comprising: repeating the feature detection on the track image using the modified search rules to label the attribute in the track image as the feature of the railway captured in the track image in response to the registration attempt failing to align the feature with the predicted track configuration context.
